

25CS307-Assignment3 Exercise Questions

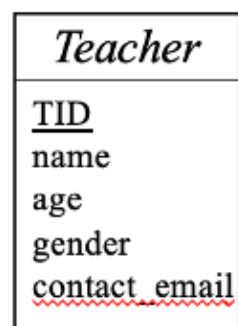
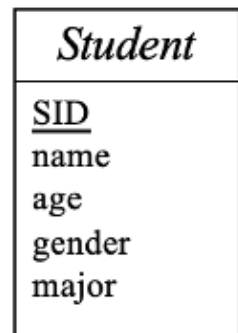
This assignment accounts for 5 points of the total score.

Question 1 (25 points):

Consider the following relation schemas when building a relational database for an online open-course website:

- **CourseVideo**(VID, title, description, release_timestamp, length)
- **Teacher**(TID, name, age, gender, contact_email)
- **Student**(SID, name, age, gender, major)

An unfinished E-R diagram is provided:



1. Complete the E-R diagram (15 points)

Besides the three schemas, please also satisfy the following requirements by adding new elements:

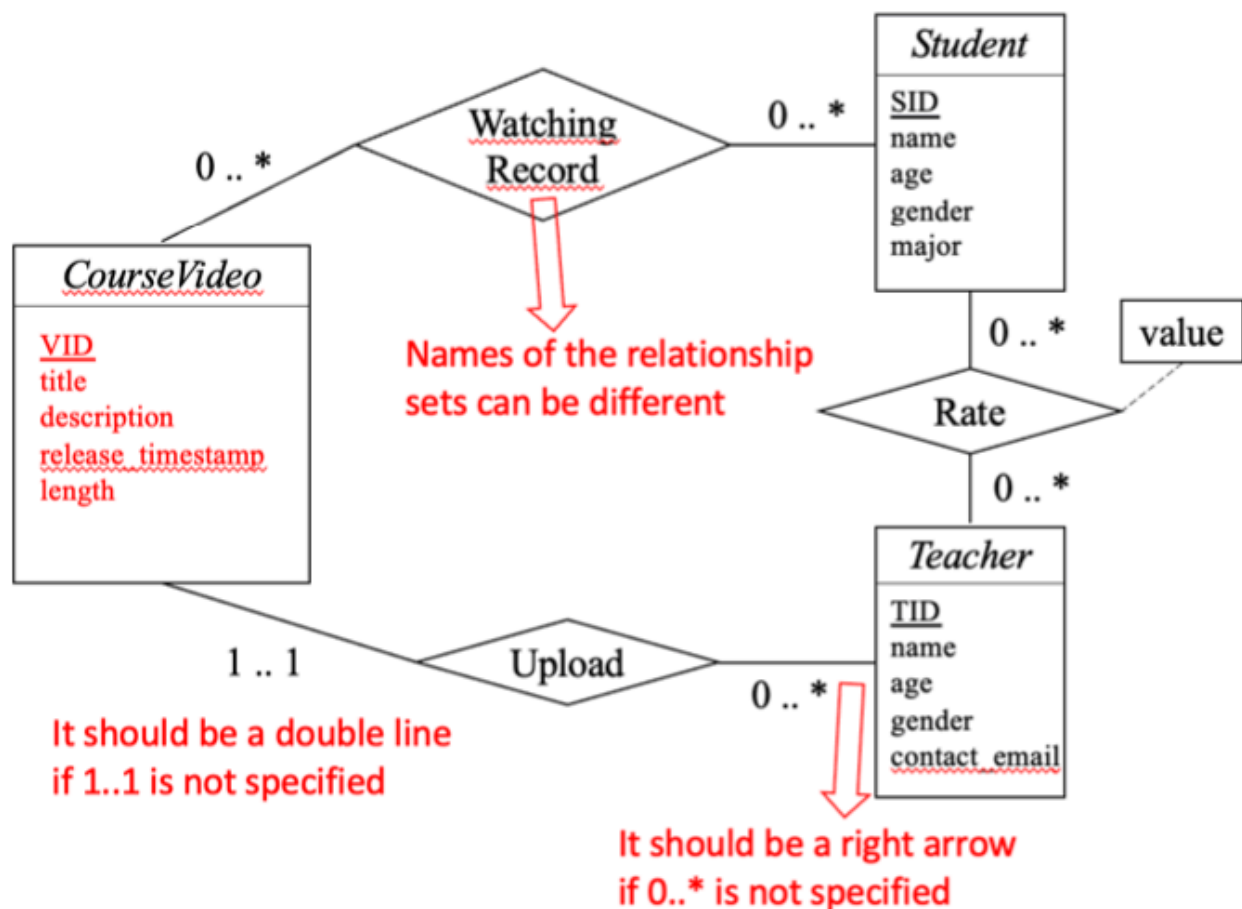
- Teachers can upload any amount of course videos. A video can only have one uploader (i.e., teacher). All videos should have an uploader. There might be occasions where some teachers

have not uploaded any video yet.

- Students can watch any video uploaded by teachers, and some students may only register accounts with no further actions. The watching records should be modeled as a relationship between students and videos. Students can rate any amount of teachers. The rating is an integer between 0 to 100.

You also need to pay attention to the following points:

- The attributes with underlines in the schemas listed above can be considered unique identifiers of the entities.
- All attributes should be illustrated inside the corresponding entity sets in the diagram. For relationship sets, please also draw the necessary attributes.
- From the requirements, please identify the cardinality constraints and participations. Primary keys, cardinality constraints, and participations should be marked in the correct format when drawing the diagram.



2. Relational Algebra (10 points)

2.1 Write relational algebra expression that returns a set containing the attribute `name` of Students whose age is larger than or equal to 20.

```
 $\pi_{name} (\sigma_{age \geq 20} (Student))$ 
```

2.2 Write relational algebra expression that returns a set containing the `video title`, `video length` and `teacher name`, to represents the courses videos have been uploaded by the teacher named "Ma".

```
 $\pi_{\text{title, length, name}} ($   
   $\sigma_{\text{name} = \text{'Ma'}} ($   
     $(\text{CourseVideo} \bowtie \text{Upload} \bowtie \text{Teacher})$   
  )  
)
```

Question 2: Normalization (25 points)

We have a relation $R(A, B, C, D)$ where A is a known superkey. A set of functional dependencies is also provided: $\{(A, C) \rightarrow B, B \rightarrow C, B \rightarrow D\}$. For the two questions below, please show the steps of proofs and decompositions. You will only receive 50% of the score if the reasoning process is missing.

1. Is (A, C) a superkey of R ? (8 points)

Answer: Yes. A is a known superkey, and any superset of a superkey is still a superkey. Thus, (A, C) is a superkey.

2. Please prove that R is not in BCNF. (8 points)

Answer: For the functional dependency $B \rightarrow C$, it holds on R . However, $B \rightarrow C$ is non-trivial, and B is not a superkey of R .

3. Decompose R into relations that satisfy BCNF. (9 points)

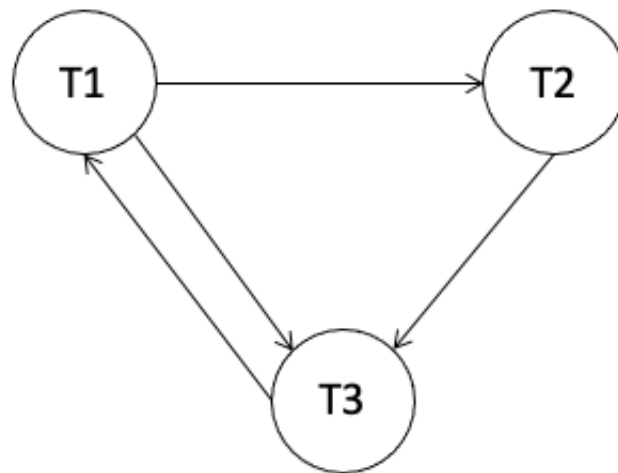
Answer:

- Since $B \rightarrow C$ violates BCNF, R can be decomposed into (B, C) and (A, B, D) .
- However, (A, B, D) still violates BCNF given $B \rightarrow D$. Thus, it can be further decomposed into (A, B) and (B, D) .
- The final decomposition is $(B, C), (A, B), (B, D)$.

Question 3: Transaction (20 points)

	T1	T2	T3
1	R(A)		
2		R(B)	
3		W(A)	
4			W(A)
5			W(B)
6	R(B)		

1. Draw the precedence graphs for the schedule above? (10 points)



2. Please explain whether the schedule is conflict serializable. (10 points)

Answer: Schedule is not conflict serializable. Its precedence graph contains a cycle.

Question 4 Query Optimization (20 points)

Suppose that we have a class enrollment system with the following three relations:

Student = (sid, sname, major, gender)

Course = (cid, cname, credit, department)

Enrollment = (sid, cid, grade, year)

Given the following SQL query, answer the questions below.

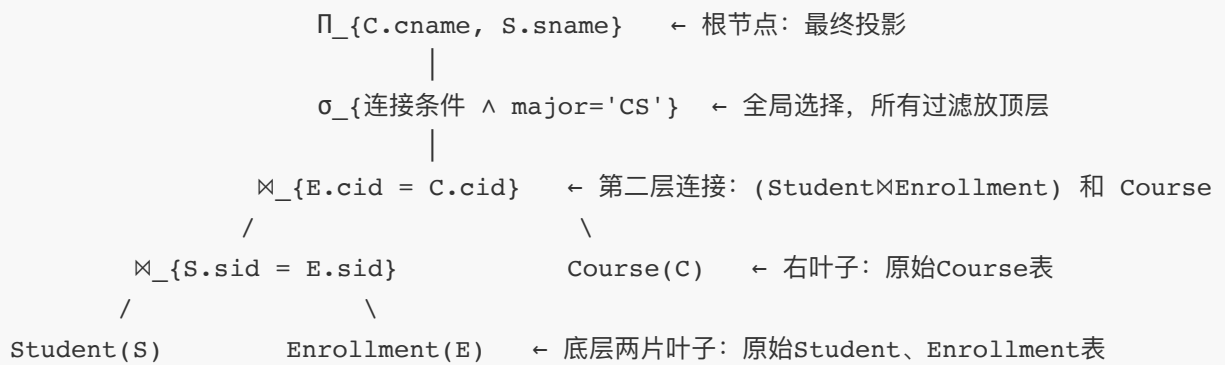
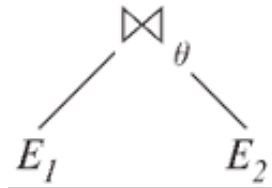
```

SELECT C.cname, S.sname FROM Student S, Enrollment E, Course C
WHERE S.sid = E.sid AND C.cid = E.cid AND S.major = 'CS'
  
```

4.1 Write down the relation-algebra expression corresponding to the above SQL statement (4 point),

$$\Pi_{C cname, S sname} \left(\sigma_{S.sid=E.sid \wedge E.cid=C.cid \wedge S.major='CS'} \left((Student\ S) \bowtie (Enrollment\ E) \bowtie (Course\ C) \right) \right)$$

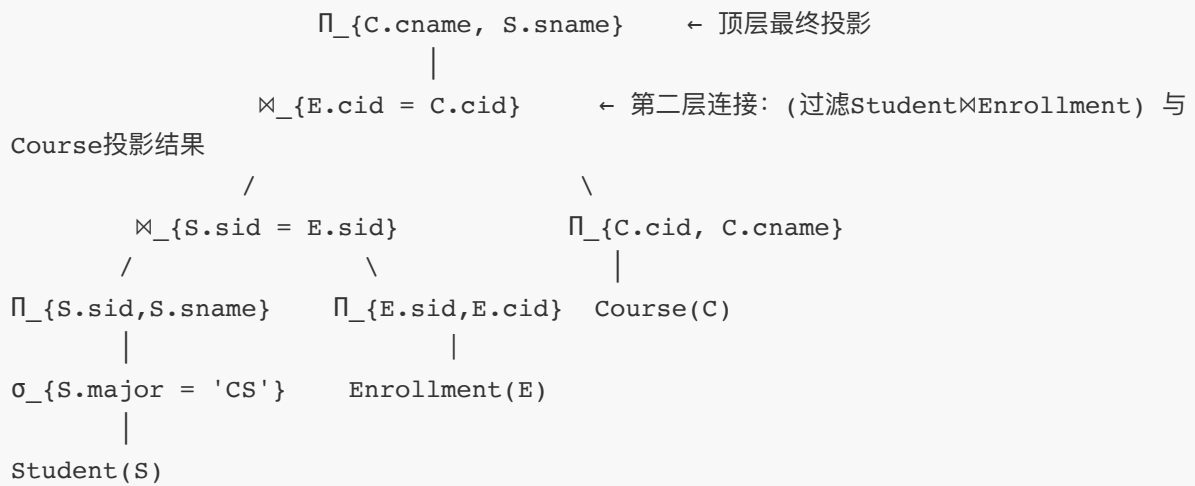
and draw an expression tree (hint: an example expression tree is shown below) corresponding to the relation-algebra expression (4 point).



4.2 Assuming that an index is built on *Student.major* and there are no other indexes, optimize the relation-algebra expression as much as possible and write down the optimized relation-algebra expression (8 points),

$$\Pi_{C.cname, S.sname} \left(\left(\Pi_{S.sid, S.sname} \left(\sigma_{S.major='CS'} (Student\ S) \right) \bowtie_{S.sid=E.sid} \Pi_{E.sid, E.cid} (Enrollment\ E) \right) \bowtie_{E.cid=C.cid} \Pi_{C.cid, C.cname} (Course\ C) \right)$$

and draw the optimized expression tree (4 points). Justify your answer.



Submission (10 points)

Submission requirements:

1. **Handwritten** assignments are required, and the question number must be marked, and the answers must be written clearly and easily understood.
2. The paper for writing assignments must be **blank A4** paper.
3. **Scanning** and uploading are required. If you upload a photo, make sure **it is clear**.